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Integrating agentic AI and digital twins for intelligent decision-making systems

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ABSTRACT

The integration of agentic artificial intelligence (agentic AI) and digital twins (DTs) enables decision-making systems that are intelligent, adaptive, and goal-oriented. This paper advances that convergence through a multilayer integration framework that organizes perception, knowledge and data management, LLM-based reasoning, learning, decision-making, action execution, and feedback adaptation into a cohesive structure. Within this framework, LLM-driven agents act as cognitive cores for context-aware planning and collaboration, while the digital twin provides a shared situational model that maintains real-time state, simulation capabilities, and policy constraints. The contributions are threefold: (i) a novel architecture that operationalizes continuous learning and closed-loop autonomy through tightly coupled agents and a DT; (ii) a use-case demonstration in power balancing for electrical grid management, where agents coordinate demand forecasting, distributed energy resources, and network constraints via a grid digital twin; and (iii) an analysis of key enablers and challenges, such as model synchronization, interpretability, safety, and cognitive load distribution, necessary for scalable and trustworthy deployment. The proposed integration offers a principled path to resilient, transparent, and data-efficient decision-making systems for cyber–physical infrastructure and provides a foundation for extending agentic AI-DT synergies to other safety-critical domains.

1. Introduction

The increasing complexity of cyber–physical systems demands intelligent decision-making frameworks that can operate autonomously, adapt dynamically, and continuously learn from their environments [1]. Two rapidly evolving paradigms, digital twins (DTs) [2] and agentic artificial intelligence (agentic AI) [3], are converging to address this challenge. DTs provide dynamic, data-driven virtual replicas of physical systems that enable real-time monitoring, prediction, and optimization [4]. Agentic AI refers to artificial intelligence that autonomously perceives system states, identifies operational goals, formulates and evaluates decision strategies, executes actions through adaptive control mechanisms, and continuously refines its performance based on feedback and changing conditions. The integration of these two technologies creates new opportunities for intelligent, self-evolving decision systems capable of optimizing complex operations in real-world environments [5,6].

Despite rapid advances in both fields, much of the literature still treats DTs as passive mirrors of assets and AI agents as isolated reasoning modules [7–9]. For example, in the grid-balancing setting of Fig. 1, intermittent sources (wind, solar) feed both the grid and utility-scale storage, while baseload plants (hydro, nuclear, thermal) anchor supply. On the demand side, heterogeneous loads (homes and commercial buildings, EV fast-charging, energy-intensive industry, and retail) pull power through the transmission network. Storage and power-to-X assets (batteries, thermal stores, electrolyzers producing hydrogen/synthetic fuels) are the pivotal buffers, absorbing surplus renewables, shifting energy from milliseconds to days, and returning it as electricity, heat, or fuels. Yet when control of these assets is siloed, situational awareness fractures across the supply-storage-demand chain, blocking online adaptation, cross-domain learning (physics ↔ markets ↔ weather ↔ behavior), and true closed-loop control. Today's operations still lean on predefined rules and periodically solved centralized optimizations that are brittle under non-stationarity (volatile demand, renewable intermittency, communication delays, model drift, unforeseen contingencies) [10,11], and the absence of a unified cognitive substrate limits explainability and hampers collaborative decisions among operators, markets, and automated controllers [12,13].

Integrating agentic AI with DT architectures addresses these limits. A continuously updated grid twin provides a physically grounded, high-fidelity state and simulation space spanning supply, storage/power-to-X, and demand, while agentic AI contributes goal-directed reasoning

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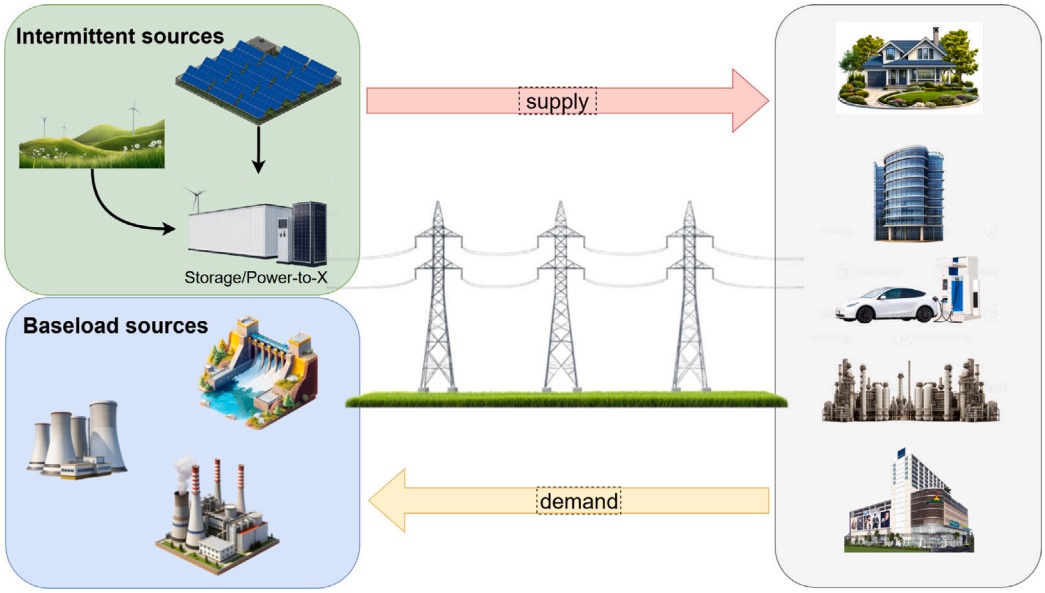

Fig. 1. Conceptual role of the grid-balancing problem as the motivating scenario for the proposed agentic AI-DT integration framework. The figure highlights why maintaining supply–demand equilibrium in a dynamic grid requires continuous perception, forecasting, and coordinated decision-making-capabilities that the integrated architecture is designed to support and enhance.

and self-improving policies. Together they enable real-time cognitive coordination: autonomous agents perceive the system through the twin, hypothesize and test balancing actions in simulation, select strategies under network and market constraints, and execute controls with immediate feedback for learning and recalibration. This fusion supports adaptive, distributed, and explainable control at grid scale (scheduling charging, dispatching storage, routing excess renewables to electrolysis, and back-feeding power when reliability or prices warrant), thereby improving robustness to variability, accelerating disturbance response, and aligning local actions with system-level objectives.

This paper proposes a multilayer integration framework that unites agentic AI and DTs to support intelligent decision-making in dynamic, high-stakes environments. The architecture introduces distinct yet interconnected layers for perception, knowledge and data management, reasoning and learning, decision-making, action execution, and feedback adaptation. Within this framework, large language model (LLM)-based agents operate as cognitive cores, which is central reasoning modules responsible for interpreting context, formulating plans, and coordinating with other agents, while the DT acts as a shared situational model that synchronizes physical states, simulations, and decision contexts.

The primary contribution of this paper is threefold:

- **A novel multilayer architectural framework integrating agentic AI and digital twin technologies**. Unlike existing hybrid AI-DT architectures that typically rely on fixed machine-learning models or rule-based controllers, the proposed framework employs LLM-driven agents as cognitive cores capable of continuous learning, contextual reasoning, and adaptive decision-making.
- **A use-case demonstration in electrical grid power balancing that operationalizes the full agentic AI-DT loop**. While previous AI-DT systems in energy management often focus on prediction or optimization in isolation, our implementation showcases how reasoning-enabled agents coordinate demand forecasting, distributed energy resources, and network constraints through a grid DT acting as a shared situational model.
- **An analysis of key enablers and unresolved challenges for scalable intelligent infrastructures**. Building on but extending beyond earlier analyses of AI-DT integration, we identify requirements specific to agentic AI, such as synchronized cognitive load distribution, interpretability of LLM-driven reasoning chains, and stable agent-DT co-evolution.

The remainder of this paper is organized as follows. Section 2 reviews related work on DT architectures, agentic AI systems, and emerging approaches to AI-DT integration. Section 3 presents the integrated conceptual and system architecture that unifies agentic AI and DTs for intelligent decision-making. Section 4 demonstrates the application of this framework to power balancing in electrical grid management as an illustrative use case. Section 5 discusses key findings, challenges, and research implications. Section 6 outlines future research directions, and Section 7 concludes the paper.

2. Related work

The integration of agentic AI and DT technologies has gained increasing attention as both paradigms mature. Each offers complementary capabilities: DTs provide a data-driven, virtualized view of physical systems, while agentic AI brings autonomy, reasoning, and adaptability. This section reviews prior research in three key areas: DT architectures, agentic AI systems, and emerging agentic AI-DT integration efforts, highlighting their evolution, limitations, and the gap addressed by this study.

2.1. Digital twin technologies

The concept of the DT originated in the manufacturing and aerospace industries as a means to create virtual counterparts of physical assets for monitoring and predictive maintenance [14]. Foundational work by Grieves [15] and NASA's [16] early implementations established the DT as a bridge between physical and digital spaces, connected through continuous data exchange. Over time, the concept has evolved from static models to dynamic, multi-domain simulations capable of reflecting the current operational state of complex systems in real time [17].

Recent studies have extended DT applications to cyber–physical and infrastructure systems, including smart manufacturing [18], transportation networks [19], and energy systems [20]. In the context of the electrical power grid, DTs have been explored for state estimation, contingency analysis, and fault prediction [21]. However, most existing grid DTs rely on deterministic optimization or rule-based control, which limits their ability to adapt to changing demand, renewable energy variability, and distributed generation [22]. Traditional DTs are excellent at representation but limited in reasoning, since they simulate but rarely decide. This shortcoming shows the need for cognitive extensions that allow DTs to autonomously learn and act within evolving operational contexts.

2.2. Agentic artificial intelligence

The recent rise of agentic AI marks a major shift from task-specific automation to goal-oriented autonomy [23]. Built upon LLMs and multimodal learning, agentic AI architectures employ iterative reasoning, tool use, and environment interaction to pursue high-level objectives with minimal human intervention. These systems embody an "agentic loop" of perception, reasoning, action, and reflection, enabling self-improvement through experience. Research on multi-agent systems (MAS) and cognitive architectures such as SOAR [24], Adaptive Control of Thought-Rational (ACT-R) [25], and more recently, LLM-based frameworks like AutoGPT [26], have contributed to the theoretical foundation for agentic AI. These developments emphasize modularity, collaboration, and adaptive learning. However, most implementations operate in digital-only environments, where agents interact within simulated or data-defined contexts, lacking a connection to live, physical systems [27]. As a result, agentic AI systems often face challenges in grounding their reasoning in real-world dynamics, data fidelity, and safety-critical constraints, areas where DTs can play a vital role.

2.3. Integration of AI and digital twin models

Integrating AI with DTs has been explored primarily through machine learning-enhanced twins, where predictive models or reinforcement learning agents optimize DT behavior [28]. Such hybrid approaches have shown promise in domains like predictive maintenance, smart manufacturing, and energy management. For example, reinforcement learning-based DTs have been used for optimal control of microgrids and dynamic load balancing [29]. However, these systems typically embed narrow AI models within DT frameworks rather than integrating agentic AI architectures capable of autonomous goal pursuit, collaboration, and self-reflection. Emerging work suggests that coupling LLM-driven reasoning agents with DTs can enable adaptive control and explainable decision-making [30]. Still, there remains a gap in defining systematic architectures that formalize how agentic AI components interact with DT representations in a closed feedback loop. Moreover, the majority of AI-DT integration efforts in power systems focus on predictive accuracy or optimization efficiency, with limited attention to cognitive adaptability, inter-agent communication, or real-time learning.

2.4. Research gap and motivation

While prior studies have advanced DT fidelity and AI autonomy independently, few have achieved a unified framework that integrates both into a cohesive, intelligent decision-making system. Current DT-based control architectures lack adaptive reasoning, whereas current agentic AI systems lack physical grounding and real-time synchronization [31]. For electrical grid management, this disconnect restricts the potential for continuous power balancing, distributed coordination, and resilience under uncertainty [32].

The insights drawn from the literature highlight persistent limitations in existing AI-DT integrations, particularly the lack of mechanisms for continuous learning, coordinated reasoning across agents, and tightly coupled cognitive-simulation loops. These gaps indicate the need for an architectural framework that systematically operationalizes agentic intelligence while leveraging the DT as a unified, real-time representation of system state. Motivated by these observations, the next section introduces a multilayer integration framework designed to address these shortcomings and support adaptive, goal-driven decision-making in complex environments.

3. Integrated conceptual and system architecture

This section presents a unified multilayer integration framework that combines agentic AI with DT to enable adaptive, goal-oriented, and physically grounded decision-making. The approach establishes a closed cognitive-cyber–physical loop connecting perception, reasoning, action, and feedback. We demonstrate the framework through power balancing in electrical grid management, a domain requiring real-time adaptability and coordination among distributed resources. Fig. 2 illustrates the high-level architecture.

3.1. Core concept

At the center of the framework is a bidirectional interaction between agentic AI systems and the DT environment. The DT is a living model of the physical grid, continuously synchronized with sensor, meter, and control data. Agentic AI agents operate as cognitive entities that interpret this model, reason about system states, and select actions aligned with operational objectives. The DT thus evolves from a passive monitor to an active cognitive ecosystem, a shared reasoning substrate where agents simulate, test, and coordinate actions before implementation. The overall system forms a continuous perception-reasoning-action–feedback loop in which decisions adapt in real time to changing conditions.

3.2. Multilayer integration framework

The architecture comprises multiple autonomous AI agents interacting with a shared DT, with a feedback and adaptation mechanism driving continuous improvement. Each agent has perception, reasoning, and execution capabilities, while the DT provides coordination, synchronization, and predictive simulation. The framework is organized into six interdependent layers:

(a) Multimodal perception layer. Interfaces the physical system, DT, and agents by aggregating heterogeneous data streams (sensors, weather, markets, and logs) into semantically consistent inputs. For power balancing it collects generation/consumption data, voltage, current, frequency indicators, renewable forecasts, and market or policy constraints.

(b) Knowledge and data layer. Serves as the cognitive memory combining structured databases, ontologies, and historical records with domain knowledge. It provides shared access to load profiles, generation forecasts, network topology, component health, and regulatory constraints for both agents and the DT.

(c) Reasoning and learning layer (LLM-based). The cognitive core of each agent. LLMs with domain-specific learning enable context-aware interpretation, anomaly detection, prediction of system behavior, hypothesis generation, and cross-agent coordination.

(d) Decision-making layer (LLM-based). Translates insights into actionable strategies by evaluating alternatives under multi-objective criteria (stability, efficiency, sustainability). In grid operations this includes dispatching storage, curtailing renewables, reconfiguring topology, or triggering demand response. Candidate actions are validated in the DT before real-world execution.

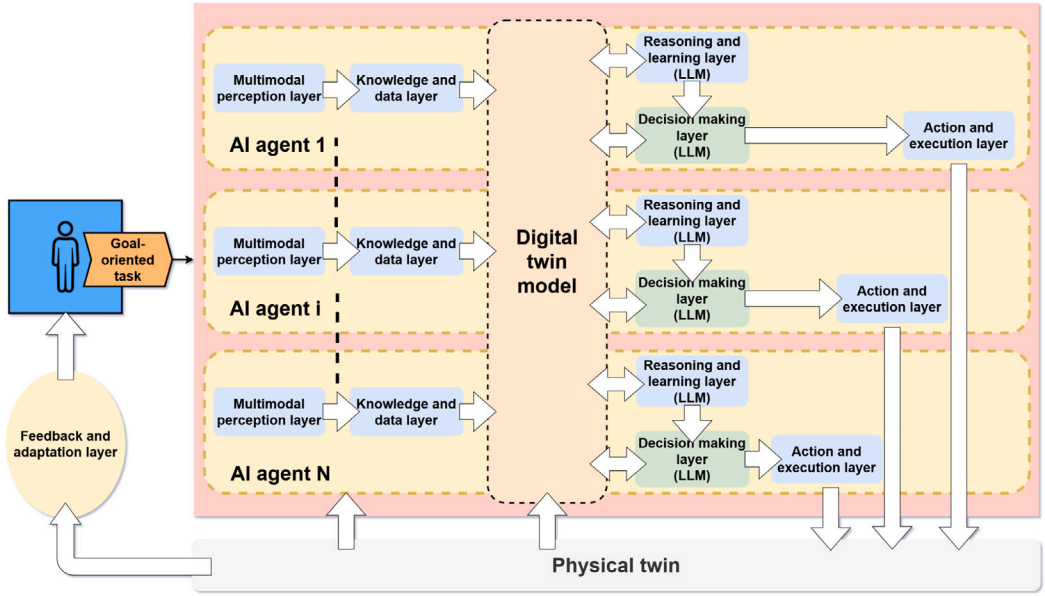

Fig. 2. Analytical structure of the proposed multilayer agentic AI-DT framework, illustrating how distinct cognitive and physical layers interact to enable continuous, goal-oriented decision-making. The figure highlights the functional rationale of each layer: perception, knowledge/data management, LLM-based reasoning, decision, action, and feedback, and shows how the DT serves as the central synchronization and simulation mechanism that grounds agent reasoning in real-time system state.

(e) Action and execution layer. Implements approved strategies through control interfaces to physical assets or operator dashboards. All commands are mirrored in the DT to ensure traceability and model synchronization, closing the perception-action loop.

(f) Feedback and adaptation layer. Aggregates outcomes and performance metrics, updating both the DT and agents' policies for continuous improvement via incremental learning, model recalibration, and cross-agent knowledge sharing.

A key rationale behind the proposed framework is its modular decomposition of perception, reasoning, decision-making, and actuation. This separation reduces system complexity by allowing each module to specialize and operate within a well-defined functional boundary. Such modularity limits the propagation of errors that often arise in monolithic AI pipelines and facilitates debugging, verification, and targeted refinement of individual components. By distributing cognitive tasks across dedicated agents rather than relying on a single, opaque model, the architecture increases interpretability and operational robustness, enabling the system to respond adaptively to dynamic conditions without sacrificing transparency.

3.3. Role of the digital twin

The DT serves as the central cognitive hub and simulation engine that unifies perception, reasoning, and control. First, as a system representation, it maintains a real-time, high-fidelity model of generation, transmission, and demand, continuously synchronized with telemetry and enriched with contextual information to reflect the current operating state. Second, as a decision sandbox, it provides a risk-free environment in which agents can execute what-if studies, validate forecasts, and stress-test candidate dispatch strategies against operational constraints before field deployment. Third, as a coordination layer, it offers a shared workspace that aligns multi-agent objectives, exposes common state and constraints, and mediates trade-offs so that local decisions remain coherent with system-level goals. Together, these functions allow the DT to ground LLM-based reasoning in physics, de-risk decision-making, and enable consistent, collective behavior across heterogeneous agents.

4. Implementation scenarios

To demonstrate the applicability of the proposed agentic AI-DT integration architecture, this section presents its implementation in the context of the grid balancing problem within an electrical power system. Grid balancing represents a complex, dynamic challenge that requires continuous coordination between diverse energy generators and heterogeneous consumers. It serves as an ideal testbed for the proposed framework, illustrating how autonomous agents, guided by a shared DT environment, can collectively perceive, reason, and act to maintain real-time equilibrium between supply and demand while adapting to operational uncertainties.

Fig. 3 depicts the grid-balancing scenario and how it is orchestrated by the proposed agentic AI-DT architecture. On the left, four heterogeneous generators (hydroelectric, geothermal, wind, and solar) feed a portfolio of consumers (households, businesses, industry, and public facilities). Each resource is paired with a local AI agent: generation-side agents supervise production and availability, while demand-side agents optimize consumption and local efficiency. Red links indicate potential power routing between generators and consumers.

On the right, the central grid-balancing agent is organized around a DT simulation core that maintains a time-synchronized, physically grounded representation of the grid. Telemetry from the supply side (generator outputs, availability, ramp rates) and the demand side (class-level loads, demand-response status) is first validated, time-aligned, and fused by the multimodal perception module. The curated streams, along with contextual information such as weather, calendar effects, and market signals, are persisted in the knowledge & data layer, which also stores equipment limits, network constraints, and historical performance indicators. Within the highlighted DT block, the system conducts what-if simulations that couple data-driven components with physics-based models of the network. The LLM-based reasoning layer orchestrates short-term load forecasting, detects anomalies by cross-checking sensor signals with the DT state estimate, and proposes hypotheses about emerging conditions (e.g., renewable drop-offs, line deratings, or demand spikes). Each hypothesis is instantiated as a scenario and propagated through the DT to generate trajectory estimates for states and key performance metrics such as power flows, reserve margins, losses, and operating cost.

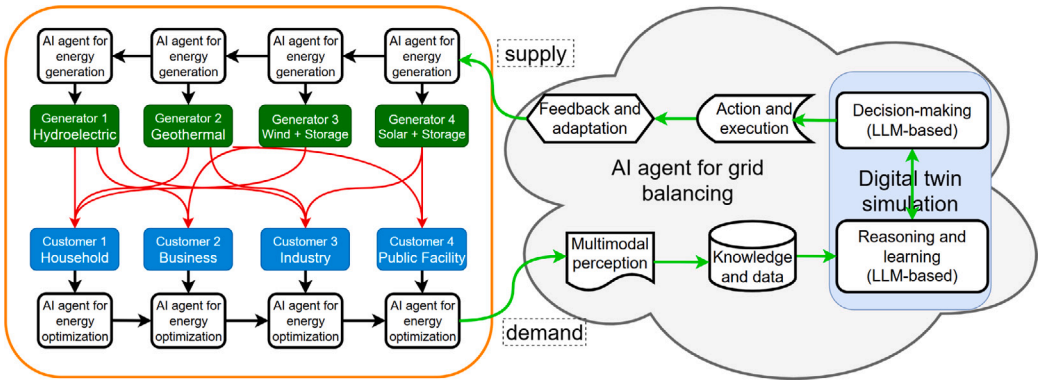

Fig. 3. Grid balancing with the proposed agentic AI-DT architecture. Local agents manage generators and consumers, while a central agent coordinates system-level balance. The DT serves as the simulation hub where LLM-based reasoning and decision-making are run, tested, and validated via what-if scenarios before execution. Perception and knowledge layers feed the DT; action and continuous feedback close the loop for stable, economical, real-time operation.

The LLM-based decision-making layer then evaluates candidate dispatch strategies under these scenarios. Strategies are screened against operational requirements including generation capacity, ramping limits, reserve targets, and aggregate transfer limits. Before any command is issued to the field, candidate actions are exercised and stress-tested inside the DT to assess feasibility and impact. The simulator verifies that supply and demand remain balanced, network constraints are respected, and operating cost and risk targets are acceptable. Only strategies that satisfy these checks are promoted to the action & execution module and translated into control directives for generators, storage, and demand-response programs. Executed actions produce new observations that are reconciled with the DT predictions. The feedback & adaptation loop continuously calibrates forecasting models and surrogate network models, updates constraint estimates, and audits performance using interpretable metrics. In this configuration, the DT is not a passive mirror but a cognitive sandbox: it grounds reasoning in physics, validates and de-risks decisions prior to deployment, and supports coordinated, resilient grid operation with continual learning.

In this simulation, the multimodal perception layer continuously acquires hourly electricity-usage measurements from four customer classes: households (C1), businesses (C2), industry (C3), and public facilities (C4), as shown in Fig. 4. The resulting time series are curated in the knowledge and data layer, where they are cleaned, time-aligned, and optionally enriched with contextual features (e.g., calendar effects, weather, special events). These processed data provide the basis for short-term load forecasting, producing 24-hour-ahead demand predictions that drive the subsequent decision-making and control actions.

The short-term load forecasting is performed in the reasoning and learning layer using a Random Forest Regressor (RFR). For each customer class $i \in \{1,2,3,4\}$ and forecast horizon $h \in \{1,\dots,24\}$, the predicted demand is

$$\hat{D}_{i,t+h} = f_{\mathrm{RFR}}(X_{i,t}),$$

where $X_{i,t}$ aggregates lagged loads, calendar indicators, weather features, and temporal encodings (hour-of-day, day-of-week). Class-level forecasts are reconciled to the system level via

$$\hat{D}_{t+h} = \sum_{i=1}^{4} \hat{D}_{i,t+h}.$$

Given the 24-hour forecasts $\{\hat{D}_{t+h}\}_{h=1}^{24}$, we solve, for each hour h, an economic dispatch that minimizes generation cost while balancing supply and demand and respecting simple capacity limits. Let P_i be the output of generator $i \in \mathcal{G}$, with linear cost C_i (€/MWh) and capacity $0 \le P_i \le P_i^{\max}$. Furthermore, let $P_{i,j}$ denote the power transmitted from generator i to customer j, where $j \in C$ represents the set of all customer nodes in the system. We include two nonnegative slack variables: U_h for unmet demand (load shedding) and S_h for surplus/spillage to keep the model always feasible, with penalties $C_U \gg C_i$ and $C_S \ge 0$. The optimization problem is given by

$$\begin{aligned}
\min_{\{P_i, U_h, S_h\}} \quad & \sum_{i\in\mathcal{G}} C_i P_i + C_U U_h + C_S S_h \\
\text{s.t.} \quad & \sum_{i\in\mathcal{G}} P_i + U_h = \hat{D}_{t+h} + S_h && \text{(supply–demand balance)} \\
& 0 \le P_i \le P_i^{\max} && \forall i \in \mathcal{G} \quad \text{(generator limits)} \\
& \sum_{i\in\mathcal{G}, j\in C} P_{i,j} \le F^{\max} && \text{(aggregate transmission limit)} \\
& \hat{D}_{t+h} \ge 0, \quad U_h \ge 0, \quad S_h \ge 0.
\end{aligned}$$

Here, $F^{\max}$ represents a simple aggregate throughput cap for the grid (e.g., a tie-line or corridor limit). The solution $\{P_i^\star, U_h^\star, S_h^\star\}$ provides the hour-h dispatch and any required curtailment or (penalized) unmet demand. These results are passed to the LLM-based decision-making layer, which converts the dispatch into structured control directives for generators (and, if applicable, storage and demand response), while the feedback and adaptation layer updates the forecaster and limits using realized outcomes.

We implement the optimization in Python (see: https://github.com/agushasan/AgenticAIDT) using a solver-agnostic linear program. Generator and transmission parameters are provided as lightweight dictionaries for clarity and reproducibility.

```
# Generator data
generators =
{
    "G1": {"cost": 40, "min": 1000, "max": 4000},  # Wind
    "G2": {"cost": 42, "min": 2000, "max": 7000},  # Hydro
    "G3": {"cost": 47, "min": 1500, "max": 8000},  # Solar
    "G4": {"cost": 73, "min": 1000, "max": 10000}, # Geothermal
}
```

Each generator entry specifies a marginal generation cost (in €/MWh) and minimum/maximum output (in MW) for the hourly dispatch. The minimum output represents a baseload commitment that must be met whenever the unit is online. In this study, wind (G1) and solar (G3) are assumed to be coupled with co-located energy storage (or contractual firming) sufficient to honor their minimum output in the hourly timescale; this abstracts away explicit state-of-charge tracking while still allowing the model to represent firmed renewables. If a purely variable-renewable case is desired, the minimum output for those units can be set to zero and curtailment handled via transmission or surplus variables.

```
# Transmission data
transmission =
{
    ("G1", "C1"): 2000, ("G1", "C2"): 4000,
    ("G2", "C2"): 5000, ("G2", "C3"): 5000,
```

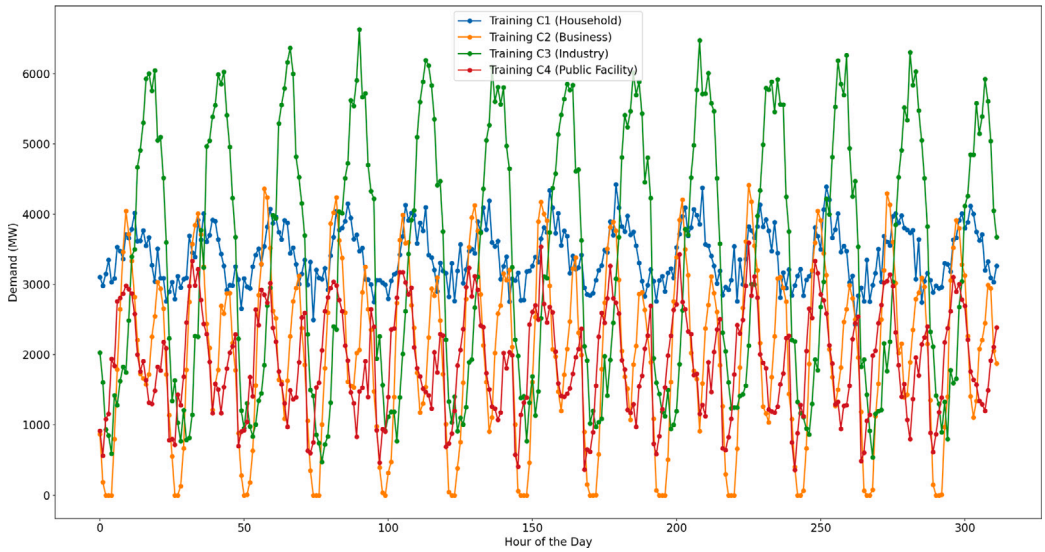

Fig. 4. Historical hourly electricity demand for four customer classes. C1: Household, C2: Business, C3: Industry, and C4: Public Facility, over a multi-day training window. The series exhibit distinct diurnal cycles and variability used to train the 24-hour demand forecaster.

```
    ("G3", "C1"): 3000, ("G3", "C3"): 4000,
    ("G4", "C2"): 5000, ("G4", "C4"): 4000,
}
```

The transmission dictionary encodes a sparse generator-to-customer capacity matrix (in MW per hour). A key `(''Gi'', ''Cj'')` indicates that power may be routed from generator `Gi` to customer class `Cj`, with the associated value giving the maximum allowable flow on that corridor; missing pairs are infeasible (capacity effectively zero). This results in a simple and transparent transportation-style network model in which flows are limited both by generator capabilities and by point-to-point transfer limits (standing in for feeder or contractual constraints).

The results in Fig. 5 illustrate the outcome of the grid-balancing optimization process performed using the proposed agentic AI-DT framework. The four subplots show the interplay between electricity demand, generation, power flow, and total cost over a 24-hour simulation period. In the top-left subplot, the demand profiles for the four customer categories: households (C1), businesses (C2), industry (C3), and public facilities (C4), exhibit distinctive diurnal patterns. Industrial (C3) demand dominates throughout the day, peaking around mid-day, while household (C1) and business (C2) loads rise during the morning and evening hours, reflecting typical work and residential activity cycles. Public facilities (C4) maintain relatively stable demand, with moderate peaks during working hours. These forecasts, provided by the reasoning and learning layer, form the target demand levels for the subsequent optimization.

The bottom-left subplot presents the dispatch schedule for the four generators: wind (G1), hydroelectric (G2), solar (G3), and geothermal (G4), as determined by the linear programming optimization in the decision-making layer. Hydroelectric power (G2) contributes the largest share of generation, ramping up during mid-day hours to compensate for the industrial demand peak. Wind (G1) and solar (G3) output fluctuate in line with their availability patterns, while geothermal (G4) provides a steady baseload contribution due to its stability and controllability. Together, these sources satisfy the aggregate demand while minimizing total generation cost and respecting the transmission capacity constraints.

The top-right subplot visualizes how the optimization allocates power among generator-consumer pairs. Each curve represents the power transferred from a generator G to a consumer C. The results show that the optimization model effectively coordinates energy routing across the network. For instance, the hydro generator (G2) serves as a major supplier, delivering substantial power to both industrial (C3) and public facility (C4) consumers during peak demand hours. In contrast, wind (G1) and solar (G3) generation are mainly directed toward household (C1) and business (C2) loads, reflecting their variability and intermittent nature. These evolving flow patterns demonstrate the system's adaptive balancing capability, redistributing generation in real time to satisfy fluctuating demand while reducing network congestion and improving overall efficiency.

The bottom-right subplot shows the temporal evolution of the total system cost resulting from the optimized dispatch. The cost curve follows the aggregate demand profile, increasing sharply during high-demand hours and decreasing as loads subside. The total generation cost peaks around the mid-day period, corresponding to maximum industrial and business activity, then gradually declines toward the evening. This behavior demonstrates the optimization's effectiveness in aligning economic efficiency with operational feasibility, achieving balance between supply, demand, and network constraints while keeping overall costs within acceptable bounds. Overall, these results confirm that the proposed architecture can adaptively allocate generation resources, minimize costs, and maintain power balance across heterogeneous energy sources and demand classes in real time.

To quantitatively assess the responsiveness of the proposed agentic AI-DT integration, we evaluated convergence time across ten distinct load conditions representing varied levels of demand fluctuation and distributed energy resource availability. For each scenario, agents iteratively coordinated through the DT to balance supply–demand mismatches and stabilize grid operating variables. The results show that the system consistently converges within a narrow time band, indicating stable and predictable performance across heterogeneous operating conditions. The small variance in convergence time suggests that the framework maintains robustness even as load profiles change, benefiting from the DT's synchronized state representation and the agents' adaptive reasoning capabilities. This consistency demonstrates that the multilayer integration supports reliable decision-making without degradation in responsiveness under different grid states.

Remark that, scaling the proposed agentic AI-DT architecture to support city- or national-level grids requires careful consideration of data fidelity, model update frequency, and computational distribution. Modern grid monitoring devices deliver updates at intervals as short as 30 s, creating high-volume data streams that must be ingested, validated, and integrated into the DT in real time. To maintain this temporal fidelity, agents must process perceptual inputs, update their

Fig. 5. Results of the grid-balancing optimization using the proposed agentic AI-DT architecture. (Top-left) Hourly demand profiles for four consumer classes: households, businesses, industry, and public facilities. (Bottom-left) Optimal power generation schedules for wind, hydroelectric, solar, and geothermal plants. (Top-right) Power flow allocations from generators to consumers derived from the optimization process. (Bottom-right) Total generation cost over time, reflecting the relationship between demand intensity and operational expenditure.

internal models, and synchronize with the DT within each reporting cycle. While the computational requirements are modest for local or regional systems, scaling to larger grids will necessitate distributed deployment of agent teams, hierarchical DT instances, and parallelized decision-making pipelines that reduce per-agent workload. Techniques such as edge-based preprocessing, asynchronous message passing, and batched simulation updates can further reduce compute demand while preserving analytic quality. These considerations highlight both the feasibility and the importance of designing scalable, resource-efficient implementations for large-scale cyber–physical infrastructures.

5. Discussion and challenges

The results of the grid balancing simulation demonstrate the potential of the proposed agentic AI-DT architecture to transform how complex cyber–physical systems are monitored, optimized, and controlled. Beyond the quantitative outcomes, several qualitative insights, operational challenges, and broader research implications emerge from this work.

The integration of agentic AI and DTs enables a shift from rule-based and centrally optimized control toward a more distributed, goal oriented, and continuously learning paradigm. In the presented case, autonomous agents equipped with forecasting, optimization, and decision capabilities effectively coordinated renewable and conventional generation to balance dynamic loads. The reasoning and learning layer (via the Random Forest Regressor and LLM-based reasoning) provided short-term demand anticipation, while the decision-making layer converted these forecasts into cost-optimal dispatch schedules that adhered to grid constraints. The DT, acting as the shared cognitive environment, allowed the agents to simulate and validate decisions before real-world execution, reducing risk and latency in the control loop. These results highlight three key advantages of the proposed framework. First, the cognitive coordination among agents allows dynamic redistribution of power across generators and consumers, improving grid stability under uncertain demand and renewable variability. Second, the adaptive learning loop enables the system to refine forecasting and optimization strategies through feedback from real-world outcomes. Third, the architecture enhances transparency and explainability, since LLM-based reasoning allows human operators to query, interpret, and justify automated decisions in natural language.

Beyond these benefits, the modular agentic structure also enhances system resilience by enabling cross-verification and redundancy across cognitive processes. The DT reinforces this resilience by functioning as a continuously updated reference model against which agent decisions can be tested prior to deployment, allowing the system to detect anomalies and mitigate cascading errors. By distributing responsibilities across multiple, independently verifiable agents rather than relying on a single monolithic AI, the architecture reduces the risk of black-box failures and strengthens the overall safety posture of the cyber–physical system.

Despite its promise, several technical and operational challenges remain:

- Data quality and synchronization: Real-time DTs depend on accurate, high-frequency data streams from sensors, smart meters, and SCADA systems. Missing, delayed, or inconsistent data can impair learning and destabilize control decisions.
- Scalability and computational cost: Training forecasting models and solving optimization problems across multiple agents requires significant computational resources. Distributed learning and hierarchical coordination strategies will be needed to scale to national or continental grid levels.
- Uncertainty management: Renewable generation, load behavior, and market conditions introduce stochasticity that is difficult to capture with deterministic optimization. Integrating probabilistic forecasting, stochastic optimization, and risk-aware decision-making remains an open area.
- Cybersecurity and trust: Connecting autonomous agents and DTs through networked infrastructure increases vulnerability to cyber threats and data tampering. Future research must incorporate secure communication protocols, anomaly detection, and verifiable decision logs.
- Human-AI collaboration: While agentic AI can operate autonomously, real-world deployment requires human oversight. Designing effective interfaces and interaction protocols that maintain situational awareness and accountability is an ongoing challenge.

Furthermore, while the framework improves transparency and coordination, its practical deployment introduces several implementation constraints directly related to interpretability and safety. Latency becomes critical as agent reasoning and DT synchronization must occur

within operational time limits; excessive latency can obscure causal traces, making it harder for operators to understand why certain actions were taken. Model drift in forecasting or reasoning modules poses another challenge, as outdated cognitive models may produce decisions that diverge from real-time system conditions, reducing both safety and explainability.

6. Future research directions

While the proposed framework demonstrates the feasibility and benefits of integrating agentic AI and DT technologies for intelligent grid balancing, several avenues merit further exploration. First, scalable multi-agent coordination and distributed optimization is needed to move from small prototypes to regional and national grids. Promising directions include consensus-based and hierarchical optimization, federated learning for model sharing without raw data exchange, and communication protocols that adapt bandwidth use to operational urgency while preserving global feasibility. Second, future work should incorporate uncertainty-aware decision making. Renewable availability, load behavior, and contingencies are inherently stochastic; therefore, probabilistic forecasting, ensemble models, and scenario-based or chance-constrained optimization are natural extensions. Such methods would allow agents to quantify and propagate uncertainty through the DT, producing risk-aware dispatch plans with explicit reliability margins. Third, tighter integration with energy markets and policy layers can align operational objectives with economic incentives and sustainability goals. Agentic AI could autonomously generate bids, negotiate demand response, and evaluate policy impacts within the twin before execution. Embedding market rules and regulatory constraints into the shared knowledge base would enable closed-loop simulations that co-optimize cost, emissions, and reliability. Fourth, advancing toward cognitive DTs and human-in-the-loop interaction will be essential for trust and accountability. Research is needed on explainable LLM-based reasoning, interactive natural-language interfaces for operators, and mechanisms for human oversight that can intercede, audit, or revise autonomous actions without breaking real-time guarantees. Fifth, the architecture should be generalized across domains. Many large-scale cyber–physical systems, such as water distribution, transportation logistics, and manufacturing, exhibit similar coupling of prediction, optimization, and control. Developing reusable ontologies, model templates, and agent roles would support transferability while respecting domain-specific constraints. Finally, ethics, security, and regulation demand sustained attention. Future work should address verifiable and auditable AI pipelines, encrypted and authenticated inter-agent communication, adversarial robustness, privacy preservation, and alignment with evolving standards. Co-design of technical and regulatory frameworks will be critical for responsible deployment of autonomous decision systems at scale.

7. Conclusion and future works

This paper proposed an integrated agentic AI-DT architecture for intelligent decision-making. The framework unifies perception, reasoning, decision-making, and feedback into a continuous cognitive-cyber–physical loop, enabling autonomous and adaptive control. Within this architecture, multimodal perception provides real-time situational awareness, while LLM-based reasoning and learning layers generate context-aware predictions and strategies. The decision-making layer formulates and solves cost-minimization problems subject to grid constraints, and the feedback layer enables continuous adaptation through real-time data and outcome evaluation. The simulation results demonstrated how the proposed architecture can forecast load, optimize generation dispatch, coordinate power flows, and reduce operational cost while maintaining grid stability. These outcomes validate the effectiveness of coupling DTs with agentic AI for distributed, goal-oriented decision-making in complex environments. Beyond the grid domain, the approach has broader implications for safety-critical cyber–physical systems. By enabling cognitive, self-adaptive, and explainable control, the integration of agentic AI and DTs represents a fundamental step toward the next generation of autonomous, resilient, and human-aligned intelligent infrastructures.

Future research may explore scaling this architecture to multi-domain environments, such as transportation, manufacturing, and smart buildings, where multiple DTs must interoperate and coordinate across shared resources. Additional directions include developing standardized protocols for agent, which include DT interaction, investigating mechanisms for cross-domain cognitive load distribution, and evaluating performance in real-world deployments. These avenues will further advance the practicality, robustness, and generalizability of agentic AI-DT systems.

CRediT authorship contribution statement

Agus Hasan: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Dong Trong Nguyen:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve readability and language. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All simulation code used in this study are openly available at the following repository: https://github.com/agushasan/AgenticAIDT.

References

[1] Thakur P, Sehgal VK. Emerging architecture for heterogeneous smart cyber-physical systems for industry 5.0. Comput Ind Eng 2021;162:107750.
[2] Tao F, Xiao B, Qi Q, Cheng J, Ji P. Digital twin modeling. J Manuf Syst 2022;64:372–89.
[3] Acharya DB, Kuppan K, Divya B. Agentic AI: Autonomous intelligence for complex goals-a comprehensive survey. IEEe Access 2025.
[4] Rasheed A, San O, Kvamsdal T. Digital twin: Values, challenges and enablers from a modeling perspective. IEEE Access 2020;8.
[5] Timms A, Langbridge A, Antonopoulos A, Mygiakis A, Voulgari E, O'Donncha F. Agentic AI for digital twin. In: AAAI conference on AI, vol. 39, 2025, p. 29703–5.
[6] Shavit Y, Agarwal S, Brundage M, Adler S, O'Keefe C, Campbell R, Lee T, Mishkin P, Eloundou T, Hickey A, et al. Practices for governing agentic AI systems. Res Pap OpenAI 2023.
[7] Alnaser AA, Maxi M, Elmousalami H. AI-powered digital twins and internet of things for smart cities and sustainable building environment. Appl Sci 2024;14(24):12056.

[8] Huang Z, Shen Y, Li J, Fey M, Brecher C. A survey on AI-driven digital twins in industry 4.0: Smart manufacturing and advanced robotics. Sensors 2021;21(19):6340.
[9] Chen S, Bekar ET, Bokrantz J, Skoogh A. AI-enhanced digital twins in maintenance: Systematic review, industrial challenges, and bridging research–practice gaps. J Manuf Syst 2025;82:678–99.
[10] Zsiborács H, Pintér G, Vincze A, Birkner Z, Baranyai NH. Grid balancing challenges illustrated by two European examples: Interactions of electric grids, photovoltaic power generation, energy storage and power generation forecasting. Energy Rep 2021;7:3805–18.
[11] Rosales-Asensio E, Diez DB, Sarmento P. Electricity balancing challenges for markets with high variable renewable generation. Renew Sustain Energy Rev 2024;189:113918.
[12] Hasan A, Rossi PS. A unified sensor and actuator fault diagnosis in digital twins for remote operations. Mech Syst Signal Process 2025;222:111778.
[13] West S, Stoll O, Meierhofer J, Züst S. Digital twin providing new opportunities for value co-creation through supporting decision-making. Appl Sci 2021;11(9):3750.
[14] Grieves M. Digital twin: manufacturing excellence through virtual factory replication. White Pap 2014;1(2014):1–7.
[15] Grieves MW. Product lifecycle management: the new paradigm for enterprises. Int J Prod Dev 2005;2(1–2):71–84.
[16] Glaessgen E, Stargel D. The digital twin paradigm for future NASA and US air force vehicles. In: 53rd AIAA/aSME/ASCE/AHS/aSC structures, structural dynamics and materials conference. 2012, p. 1818.
[17] Hasan A. Online parameter estimation in digital twins for real-time condition monitoring. IEEE Access 2025;13.
[18] Kritzinger W, Karner M, Traar G, Henjes J, Sihn W. Digital twin in manufacturing: A categorical literature review and classification. Ifac-PapersOnline 2018;51(11):1016–22.
[19] Gao Y, Qian S, Li Z, Wang P, Wang F, He Q. Digital twin and its application in transportation infrastructure. In: IEEE international conference on digital twins and parallel intelligence. 2021, p. 298–301.
[20] Haghshenas A, Hasan A, Osen O, Mikalsen ET. Predictive digital twin for offshore wind farms. Energy Inform 2023;6(1):1.
[21] Jain P, Poon J, Singh JP, Spanos C, Sanders SR, Panda SK. A digital twin approach for fault diagnosis in distributed photovoltaic systems. IEEE Trans Power Electron 2019;35(1):940–56.
[22] Chang L, Zhang L, Fu C, Chen Y-W. Transparent digital twin for output control using belief rule base. IEEE Trans Cybern 2021;52(10):10364–78.
[23] Murugesan S. The rise of agentic AI: implications, concerns, and the path forward. IEEE Intell Syst 2025;40(2):8–14.
[24] Laird JE. The soar cognitive architecture. MIT Press; 2019.
[25] Ritter FE, Tehranchi F, Oury JD. ACT-r: A cognitive architecture for modeling cognition. Wiley Interdiscip Rev: Cogn Sci 2019;10(3):e1488.
[26] Yang H, Yue S, He Y. Auto-GPT for online decision making: Benchmarks and additional opinions. 2023, arXiv preprint arXiv:2306.02224.
[27] Nicoletti B, Appolloni A. A digital twin framework for enhancing human–agentic AI–machine collaboration. J Intell Manuf 2025;1–17.
[28] Zhong D, Xia Z, Zhu Y, Duan J. Overview of predictive maintenance based on digital twin technology. Heliyon 2023;9(4).
[29] Pan M, Xing Q, Chai Z, Zhao H, Sun Q, Duan D. Real-time digital twin machine learning-based cost minimization model for renewable-based microgrids considering uncertainty. Sol Energy 2023;250.
[30] Yang H, Siew M, Joe-Wong C. An LLM-based digital twin for optimizing human-in-the loop systems. In: IEEE international workshop on foundation models for cyber-physical systems & ioT. 2024, p. 26–31.
[31] Fuller A, Fan Z, Day C, Barlow C. Digital twin: enabling technologies, challenges and open research. IEEE Access 2020;8:108952–71.
[32] Rahmani-Sane G, Azad S, Ameli MT, Haghani S. The applications of artificial intelligence and digital twin in power systems: An in-depth review. IEEE Access 2025.